# BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

**WEEKLY PROGRESS REPORT (08/06/2026 – 14/06/2026)**

**WEEK 4**

| AUGEMENTED REALITY BASED VIRTUAL CLOTHING TRY-ONS | 3D VR Integration |
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| **PROJECT DESCRIPTION :** | AN ANDROID APP THAT CREATES A 3D BODY MODEL FROM A USER PHOTO AND VIRTUALLY DRAPES SELECTED CLOTHING ITEMS ONTO IT |
| **GROUP MEMBER :** | ANANT JAIN, KARMVEER KUMAR, VISHVENDRA |
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| **08/06/2026** | **Backend AI Pipeline Restoration & PyTorch Compatibility**<br><br>• **Fixed the 4D-Humans (HMR 2.0) model initialization failure — The HMR 2.0 checkpoint was failing to load due to PyTorch 2.6+ changing the default weights_only parameter to True. Implemented a robust torch.load monkeypatch with automatic fallback (weights_only=False) and file stream pointer reset (seek(0)), along with NumPy compatibility shims for legacy SMPL pickle files. Successfully initialized HMR 2.0 on CUDA GPU.**<br><br>• **Resolved server startup conflicts and environment issues — Fixed an EGL/OpenGL crash caused by MediaPipe initializing before PyTorch by implementing lazy loading in pose_service.py. Installed missing dependencies (chumpy, webdataset, scikit-image), downgraded setuptools for Detectron2 compatibility, and configured WSL2 mirrored networking to eliminate port proxy conflicts. Verified the FastAPI server starts cleanly on port 8000.** |
| **09/06/2026** | **3D Model Quality Improvements & Texture Projection**<br><br>• **Implemented Vertex Color Projection to texture the 3D mesh — The raw HMR 2.0 output was a blank "clay" model. Used the AI-predicted camera parameters (focal length, principal point) to project each 3D vertex back onto the original 2D photo using perspective projection equations, sampling the person's clothing and skin colors directly from the image pixels and permanently painting them onto the 3D mesh vertices as RGBA colors.**<br><br>• **Eliminated background color bleeding with AI pre-processing — Vertices at the edges of the body were incorrectly sampling background pixels (walls, trees), causing color artifacts on the 3D model. Integrated a Deep Learning Background Removal step that strips all non-human pixels before the projection runs. Also fixed the coordinate system alignment (180° X-axis rotation) for proper rendering in Android's Filament engine, and added body measurement extraction (Chest, Waist, Hip, Height) from SMPL shape coefficients.** |

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| **10/06/2026** | **VR (Virtual Reality) Integration — First Version**<br><ul><li>**Built a complete stereoscopic VR viewer with head tracking — Created VRScreen.kt with two independent SceneView 3D viewports rendered side-by-side for stereoscopic depth perception (IPD offset ±0.005f).<br>Developed HeadTracker.kt using the device's Rotation Vector sensor (gyroscope + accelerometer fusion) at 60Hz with a low-pass filter for smooth tracking. Applied critical axis remapping (AXIS_Y → AXIS_MINUS_X) to account for the phone being placed sideways in the Irusu VR headset.**</li><li>**Implemented barrel distortion correction and Bluetooth remote control — Created a custom BarrelShape clipping mask using Quadratic Bézier Curves to counteract the pincushion distortion from the VR headset lenses. Integrated Bluetooth VR remote support — Joystick UP/DOWN controls camera Z-distance (zoom), Joystick LEFT/RIGHT rotates the model around Y-axis, and the Action Button resets all transformations and recenters the gyroscope.**</li></ul> |
| **11/06/2026** | **End-to-End Integration Testing, Crash Fixes & Demo**<br><ul><li>**Fixed critical crashes and optimized the data pipeline — Resolved a NullPointerException crash when transitioning to VR mode by rewriting the SceneView engine lifecycle management with remember(engine) blocks. Optimized model loading by passing the locally cached .glb file path directly to the VR screen instead of re-downloading from the network URL, making VR mode load instantly.**</li><li>**Tuned the VR experience and delivered final demonstration — Calibrated IPD from 0.032f down to 0.005f to eliminate double vision, tuned head tracking sensitivity to 1:1 mapping with ±45° pitch clamping, and fixed the yaw axis inversion for natural VR movement. Successfully demonstrated the complete end-to-end flow on the physical Vivo phone + Irusu VR headset: photo upload → AI 3D reconstruction → interactive preview → immersive VR viewing with head tracking and remote control.**</li></ul> |
| **12/06/2026** | **All members were absent** |

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| **13/06/2026** | **Holiday (Saturday)** |
| **14/06/2026** | **Holiday (Sunday)** |

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| **Next Week Plan** | **WebXR Development & Integration with 3D Digital Twin**<br><br>• **Goal:** Replace the current native Android VR implementation (SceneView/Filament) with a WebXR-based VR viewer using A-Frame / Three.js, loaded inside an Android WebView. This approach provides better cross-platform compatibility, easier maintenance, and access to the mature WebXR ecosystem for future features like hand tracking and room-scale VR. |

**REFERENCE:**


- https://shubham-goel.github.io/4dhumans/
- https://github.com/shubham-goel/4D-Humans
- https://smpl.is.tue.mpg.de/
- https://github.com/facebookresearch/segment-anything
- https://developers.google.com/mediapipe/solutions/vision/pose_landmarker
- https://github.com/google/filament